

IP ADDRESS and MAC ADDRESS .

1. What is an IP Address?

An IP Address (Internet Protocol Address) is a logical address assigned to a device for communication across networks.

Features:

- Works at Layer 3 (Network Layer) of the OSI Model.
- Used for routing data between different networks.
- Can be changed manually or automatically using DHCP.
- Unique within a network.

Example:

IPv4 : 192.168.1.10

IPv6 : 2001:db8::1

Real-World Example:

If you send a parcel to another city, the home address written on the parcel is like an IP address because it identifies the destination network and location.

2. What is a MAC Address?

A MAC (Media Access Control) Address is a physical address permanently assigned to a Network Interface Card (NIC) by the manufacturer.

Features:

- Works at Layer 2 (Data Link Layer).
- Used for communication within the same LAN.
- Usually cannot be changed permanently.
- Every NIC has a unique MAC address.

Example:

00:1A:2B:3C:4D:5E AA-BB-CC-DD-EE-FF

Real-World Example:

A person's name is like a MAC address. Even if they move to another city (change IP), their name remains the same.

OSI LAYER MAPPING

Address Type	OSI Layer	Purpose
MAC Address	Layer 2	Local Delivery
IP Address	Layer 3	End-to-End Delivery

IP vs MAC COMPARISON

Feature	IP Address	MAC Address
Full Form	Internet Protocol Address	Media Access Control Address
Layer	Layer 3	Layer 2
Type	Logical Address	Physical Address
Assigned By	Admin / DHCP	Manufacturer
Length	IPv4 = 32-bit, IPv6 = 128-bit	48-bit
Changeable	Yes	Normally No
Used For	Routing	Local Communication
Scope	Network to Network	Device to Device
Example	192.168.1.10	00:1A:2B:3C:4D:5E

HOW IP AND MAC WORK TOGETHER

Suppose:

PC1 IP : 192.168.1.10 MAC : AA-AA-AA-AA-AA-AA
PC2 IP : 192.168.1.20 MAC : BB-BB-BB-BB-BB-BB

Step 1: PC1 wants to send data to PC2.

PC1 knows:

Destination IP = 192.168.1.20

Step 2: PC1 checks its ARP table.

If MAC is unknown:

Who has 192.168.1.20? Tell 192.168.1.10

Step 3: PC2 replies:

192.168.1.20 is BB-BB-BB-BB-BB-BB

Step 4: PC1 creates a frame:

Source MAC : AA-AA-AA-AA-AA-AA Destination MAC : BB-BB-BB-BB-BB-BB

Source IP : 192.168.1.10 Destination IP : 192.168.1.20

Step 5: Switch forwards the frame using MAC Address.

COMMUNICATION ACROSS DIFFERENT NETWORKS

Example:

PC1 IP : 192.168.1.10

Gateway IP : 192.168.1.1

Server IP : 10.10.10.10

PC1 wants to reach:

10.10.10.10

Since the destination is in another network:

1. PC1 sends the packet to the Default Gateway.
2. Destination IP remains 10.10.10.10.
3. Destination MAC becomes the Router's MAC Address.
4. Router changes the Layer 2 header and forwards the packet.

Important CCNA Point:

IP Address remains the same from source to destination.

MAC Address changes at every hop.

ARP (ADDRESS RESOLUTION PROTOCOL)

Purpose:

IP Address → MAC Address

Command:

arp -a

Example Output:

Internet Address Physical Address 192.168.1.1 00-11-22-33-44-55 192.168.1.20 AA-BB-CC-DD-EE-FF

INTERVIEW QUESTIONS

Q1. Why do we need both IP and MAC addresses?

Answer: MAC Address identifies a device inside the local network, while IP Address identifies the device across different networks. IP is used for routing and MAC is used for local delivery.

Q2. Which address does a switch use?

Answer: Switch uses MAC Address.

Q3. Which address does a router use?

Answer: Router uses IP Address.

Q4. Does MAC Address change during transmission?

Answer: Yes, MAC Address changes at every hop.

Q5. Does IP Address change during transmission?

Answer: Normally No. Source and Destination IP remain the same from source to destination.

CCNA EXAM SHORTCUT

Switch = MAC Address = Layer 2

Router = IP Address = Layer 3

ARP = IP → MAC

MAC Changes Every Hop

IP Remains Same End-to-End

IP Address = Where the device is located

MAC Address = Which device it is